For example, $\begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}, \begin{pmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{pmatrix}, \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}$

In general, $A = (a_{ij})_{m\times m}$ is said to be a scalar matrix if

$$a_{ij} = \begin{cases} 0 & when \ \ i \neq j \\ k & when \ \ i = j \end{cases} \text{ where } k \text{ is constant.}$$

6. Identity (or) Unit Matrix

A square matrix in which elements in the leading diagonal are all "1" and rest are all zero is called an identity matrix or unit matrix.

Thus, the square matrix $A = (a_{ij})$ is an identity matrix if $a_{ij} = \begin{cases} 1 & if \ \ i = j \\ 0 & if \ \ i \neq j \end{cases}$

A unit matrix of order n is written as I_n.

$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ are identity matrices of order 2 and 3 respectively.

7. Zero matrix (or) null matrix

A matrix is said to be a zero matrix or null matrix if all its elements are zero.

For example, $(0), \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ are all zero matrices of order 1×1, 2×2 and 3×3 but of different orders. We denote zero matrix of order $n\times n$ by O_n.

$\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is a zero matrix of the order 2×3.

8. Transpose of a matrix

The matrix which is obtained by interchanging the elements in rows and columns of the given matrix A is called transpose of A and is denoted by A^T.

For example,

(a) If $A = \begin{pmatrix} 5 & 3 & -1 \\ 2 & 8 & 9 \\ -4 & 7 & 5 \end{pmatrix}_{3\times3}$ then $A^T = \begin{pmatrix} 5 & 2 & -4 \\ 3 & 8 & 7 \\ -1 & 9 & 5 \end{pmatrix}_{3\times3}$

(b) If $B = \begin{pmatrix} 1 & 5 \\ 8 & 9 \\ 4 & 3 \end{pmatrix}_{3\times2}$ then $B^T = \begin{pmatrix} 1 & 8 & 4 \\ 5 & 9 & 3 \end{pmatrix}_{2\times3}$

If order of A is $m\times n$ then order of A^T is $n\times m$.

We note that $(A^T)^T = A$.

10th_Maths_Chapter 3_English.indd 141 12/23/2024 5:34:22 PM